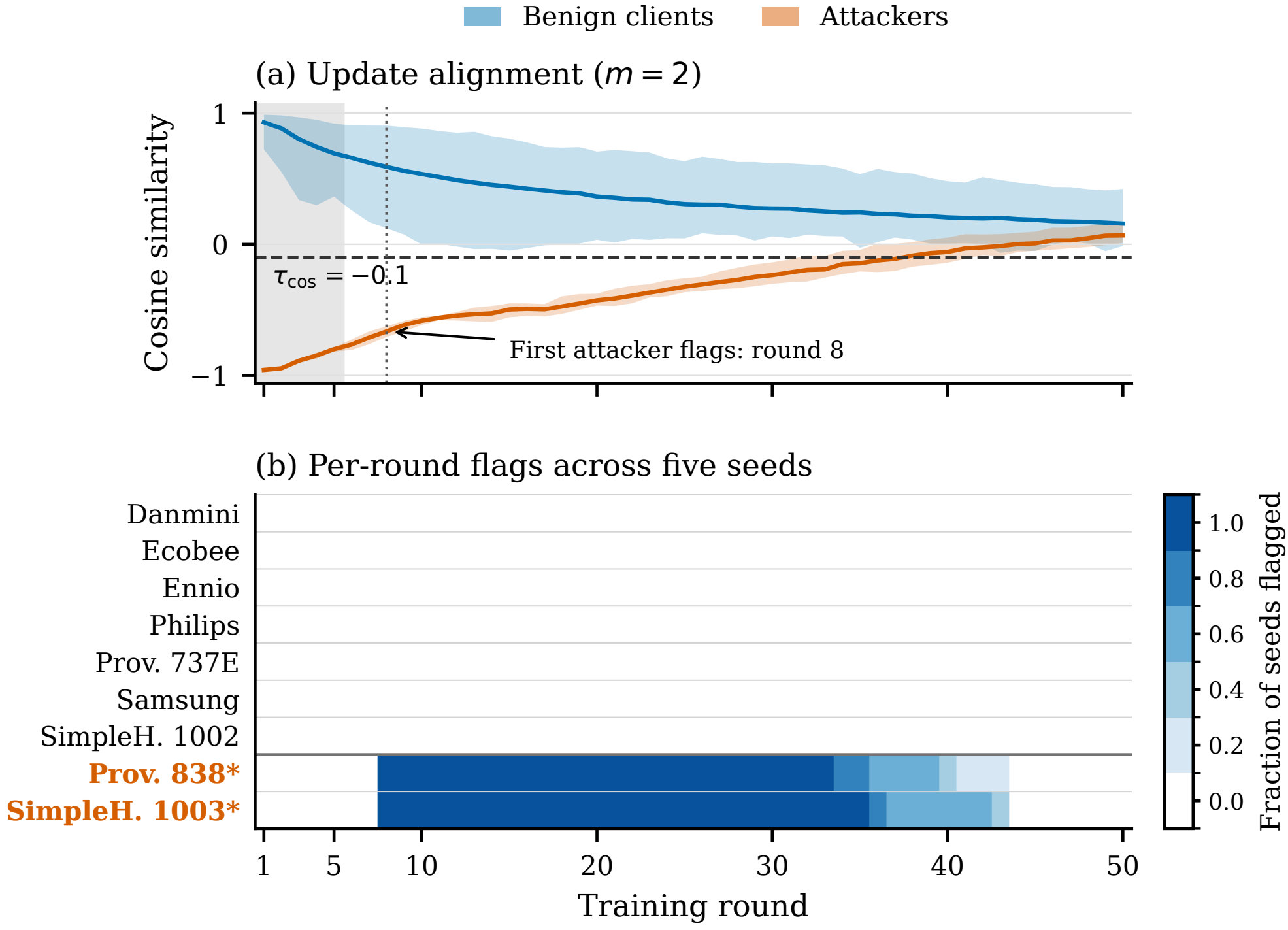
Benign clients
Attackers
(a) Update alignment ($m = 2$)
Cosine similarity
1
0
−1
$\tau_{\cos} = -0.1$
First attacker flags: round 8
(b) Per-round flags across five seeds
Danmini
Ecobee
Ennio
Philips
Prov. 737E
Samsung
SimpleH. 1002
Prov. 838*
SimpleH. 1003*
1
5
10
20
30
40
50
Training round
1.0
0.8
0.6
0.4
0.2
0.0
Fraction of seeds flagged